
ALGEBRA-SUMMER

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Contents

0.1	Intro	3
1	Groups	5
1.1	Intro	5
1.1.1	Exercises	5
1.2	Dihedral Groups	6
1.2.1	Symmetry Group for an Equilateral Triangle	6
1.2.2	Symmetry Group for a Square	8
1.2.3	Symmetry Group of regular n-gon	9
1.2.4	Exercises	10
1.3	Symmetric Groups	11
1.3.1	Cycle Notation	12
1.3.2	Exercises	15
1.4	General Results on Groups	17
1.4.1	Subgroup Lattice	18
1.4.2	Exercises	21
1.5	Cyclic Groups	22
1.5.1	Exercises	26
1.6	Homomorphism and Isomorphism	27
1.6.1	Exercises	31
1.7	Cosets	32
1.7.1	Exercises	34
1.8	Lagrange's Theorem	35
1.8.1	Exercises	37
1.9	Normal Subgroups and Quotient Groups	38
1.9.1	Exercises	39
1.10	The Alternating Group	40
1.10.1	Exercises	43
1.11	External and Internal Direct Products	44
1.12	First Isomorphism Theorem of Groups	46
1.12.1	Exercises	48
1.13	More Isomorphism Theorems	50
1.13.1	Exercises	51
1.14	Fundamental Theorem of Abelian Groups	52
1.14.1	Exercises	56

0.1 Intro

Theorem 0.1.1 (Division Algoritm). Suppose $a, b \in \mathbb{Z}$ s.t. $b > 0$. Then there are unique $q, r \in \mathbb{Z}$ s.t. $a = bq + r$ and $0 \leq r < b$

Proof. Let $a, b \in \mathbb{Z}$ s.t. $b > 0$. Consider $A = \{a - bx \mid x \in \mathbb{Z}\} \cap \mathbb{N}$

By the Well Ordering Principle since A has all non-negative terms it has a minimum lets call this r and call the associated $x \in \mathbb{Z}$, q .

Thus, $r = a - bq \implies a = bq + r$.

For contradiction, assume $r \geq b$.

$\implies r = b + m$ s.t. $m \geq 0$

$\implies b + m = a - bq \implies m = a - b(q + 1) \in A \implies m < r = \text{minimum} \implies \Leftarrow$ Thus $0 \leq r < b$

Fix $q_1, q_2, r_1, r_2 \in \mathbb{Z}$ s.t. $a = bq_1 + r_1 = bq_2 + r_2$ and $0 \leq r_1, r_2 < b$. WLOG let $r_1 \geq r_2$

$\implies bq_1 + r_1 = bq_2 + r_2 \implies r_1 - r_2 = b(q_2 - q_1) \implies r_1 - r_2$ is a multiple of b

Since $0 \leq r_1, r_2 < b \implies 0 \leq r_1 - r_2 < b$

Since $r_1 - r_2$ is strictly less than b the only multiple of b it could be is 0 thus $r_1 = r_2$

$\implies q_1 = q_2$ □

Lemma 0.1.2. (Bezout's Lemma) Suppose $a, b \in \mathbb{Z} \setminus \{0\}$. Then there are $x, y \in \mathbb{Z}$ s.t. $ax + by = \gcd(a, b)$.

Proof. Let $a, b \in \mathbb{Z} \setminus \{0\}$. Consider $A = \{ax + by \mid x, y \in \mathbb{Z}\} \cap \mathbb{N}$

By the Well Ordering Principle since A has all non-negative terms it has a minimum lets call this d and call the associated $x, y \in \mathbb{Z}$, u and v .

By Theorem 0.1.1 $\exists q, r \in \mathbb{Z}$ s.t. $a = dq + r$ and $0 \leq r < d$

Case 1: $r = 0 \implies d \mid a$

Case 2: $0 < r < d \implies r = a - dq \implies r = a - (au + bv)q = a(1 - uq) + b(-vq) \in A$

$\implies r < d = \text{minimum} \implies \Leftarrow$

Thus $d \mid a$ and similarly $d \mid b$

Let $c \in \mathbb{N}$ s.t. $c \mid a$ and $c \mid b$

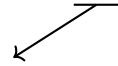
$\implies a = cm$ and $b = cn$ for $m, n \in \mathbb{Z}$

$d = au + bv = c(mu) + c(mv) = c(mu + mv) \implies c \mid d \implies d = \gcd(a, b)$ □

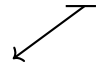
Algorithm 0.1.3. (Euclidean Algorithm)

Find $\gcd(272, 571)$ and $x, y \in \mathbb{Z}$ s.t. $272x + 571y = \gcd(272, 571)$

$$571 = 272 \cdot 2 + 27 \quad (1)$$



$$272 = 27 \cdot 10 + 2 \quad (2)$$



$$27 = 2 \cdot 13 + 1 \quad (3)$$

Last term before remainder is zero is gcd

$$\gcd(272, 571) = 1$$

$$2 = 272 - 27 \cdot 10$$

Use equation (2)

$$= 272(21) - 571(10)$$

$$1 = (571 - 272 \cdot 2) - 13(272(21) - 571)$$

Use equation (3) with (2)

$$= 571(131) + 272(-275)$$

1 Groups

1.1 Intro

Definition 1.1.1. A group is a set G together with a binary operation $*$: $G \times G \rightarrow G$ s.t.

- **(Closure)** For all $a, b \in G$ we have $a * b \in G$
- **(Identity)** There is $e \in G$ s.t. for all $a \in G$ $a * e = e * a = a$
- **(Inverses)** For all $a \in G$ there is an $a^{-1} \in G$ s.t. $a * a^{-1} = a^{-1} * a = e$
- **(Associativity)** For all $a, b, c \in G$ we have $a * (b * c) = (a * b) * c$

Definition 1.1.2. Given a group G and $g \in G$ we say the order of g is $n \in \mathbb{N}$ if n is the minimum value s.t. $g^n = g * g * \dots * g = e$
Write as: $\text{ord}(g) = |g| = n$

1.1.1 Exercises

Example 1.1.3. Let $G = \mathbb{R} \setminus \{-1\}$ and define $a * b = a + b + ab$. Show $(G, *)$ is an Abelian group

- (Abelian) For all $a, b \in G$ $a * b = a + b + ab = b + a + ba = b * a$
- (Closure) $\checkmark$
- (Identity) For all $a \in G$ $a * 0 = 0 * a = 0$
- (Inverses) For all $a \in G$ let $b = \frac{-a}{a+1}$.

$$\text{Thus, } a * b = b * a = a + b + ab = a + b(a+1) = \frac{-a}{a+1}(a+1) = a - a = 0$$

- (Associativity) For all $a, b, c \in G$

$$a * (b * c) = a * (b + c + bc) = a + b + c + bc + ab + ac + abc = (a + b + ab) * c = (a * b) * c$$

Therefore, G is an Abelian group

Example 1.1.4. Find all $x \in \mathbb{Z}$ s.t. $5x \equiv 4 \pmod{9}$

$$5x \equiv 4 \pmod{9}$$

$$x \equiv 8 \pmod{9}$$

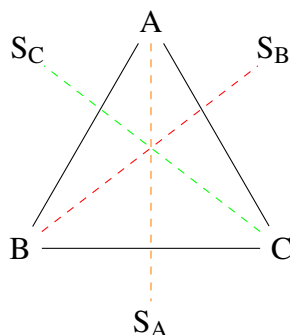
$$x = 9n + 8$$

$$2 * 5 = 10 \equiv 1 \pmod{9}$$

$$n \in \mathbb{Z}$$

1.2 Dihedral Groups

1.2.1 Rigid Motions of an Equilateral Triangle



Rotations

$r = 120^\circ$ rotation

$r^2 = 240^\circ$ rotation

$r^3 = 360^\circ$ rotation $= e = 0^\circ$ rotation

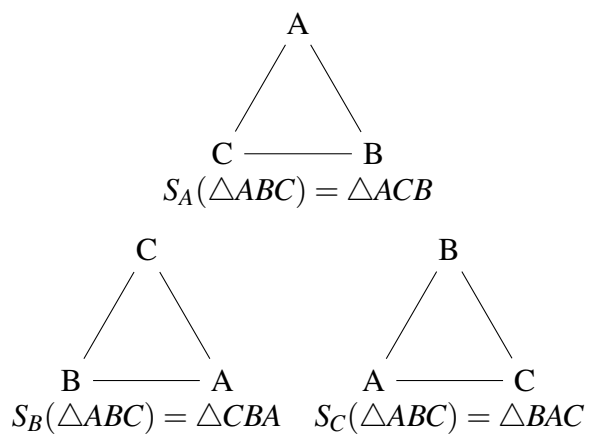
Reflections

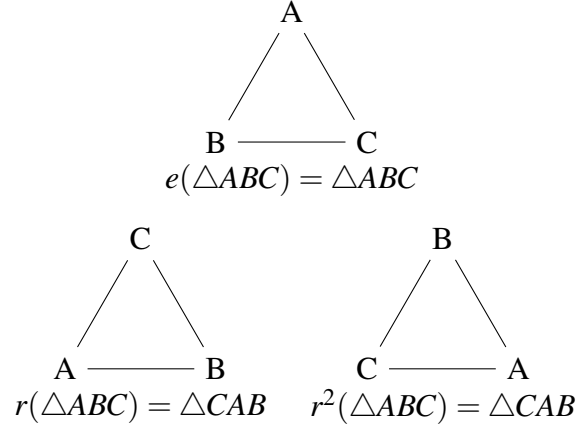
$S_A : \triangle ABC \rightarrow \triangle ACB$

$S_B : \triangle ABC \rightarrow \triangle CBA$

$S_C : \triangle ABC \rightarrow \triangle BAC$

Transformation Diagrams





Operations

$$\begin{aligned}
 r \circ S_A(\triangle ABC) &= r(\triangle ACB) = \triangle BAC = S_C(\triangle ABC) & r \circ S_A &= S_C \\
 r^2 \circ S_A(\triangle ABC) &= r^2(\triangle ACB) = \triangle CBA = S_B(\triangle ABC) & r^2 \circ S_A &= S_B
 \end{aligned}$$

Now we only need S_A so lets call this S

Thus we only need S, r, r^2, e

Also $S^2 = r^3 = e$

How S, r, r^2 Commute

$$\begin{aligned}
 r \circ S(\triangle ABC) &= r(\triangle ACB) = \triangle BAC = S \circ r^2 \\
 r^2 \circ S(\triangle ABC) &= r^2(\triangle ACB) = \triangle CBA = S \circ r
 \end{aligned}$$

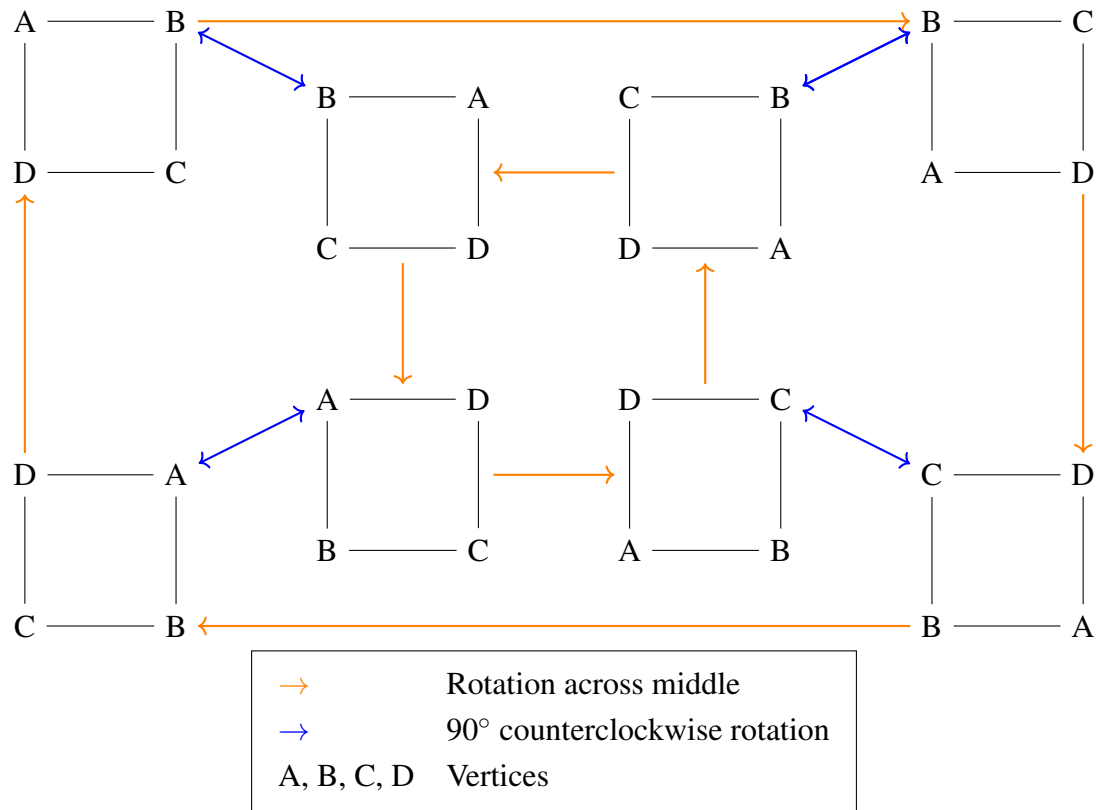
Definition 1.2.1. The group of symmetries of an equilateral triangle is $D_3 = \{e, r, r^2, S, Sr, Sr^2\}$, where $r^3 = S^2 = e$, $rS = Sr^2$

Dihedral group of order $2 \cdot 3 = 6$

Compact Notation $D_3 = \langle r, s^1 \mid r^3 = S^2 = e, rS = Sr^2 \rangle$

¹The generators of D_3 meaning you can create all other operations from them

1.2.2 Symmetry Group for a Square



Rotations

$r = 90^\circ$ rotation

$r^2 = 180^\circ$ rotation

$r^3 = 270^\circ$ rotation

$r^4 = 360^\circ$ rotation = $e = 0^\circ$ rotation

Reflections

$$s : \square ABCD \rightarrow \square BCDA$$

How does s, r Commute

From the diagram we see that $sr = r^3s \implies rs = sr^3$

Definition 1.2.2. The group of symmetries of a square is $D_4 = \{e, r, r^2, r^3, s, sr, sr^2, sr^3\}$

Dihedral group of order $2 \cdot 4 = 8$

$$D_4 = \langle s, r \mid r^4 = s^2 = e, sr = r^3s \rangle$$

1.2.3 Symmetry Group of regular n-gon

Rotation: $r = \frac{360^\circ}{n} \circlearrowleft (e, r, r^2, \dots, r^{n-1})$

Reflection: $s = \text{reflection across middle } (s, sr, sr^2, \dots, sr^{n-1})$

Claim: $rs = sr^{n-1}$.

Proof. We only need 3 points since our shape is defined by the order of those 3 points

$$r \circ s(n, 1, 2) = r(2, 1, n) = (1, n, n-1)$$

$$s \circ r^{n-1}(n, 1, 2) = s(n-1, n, 1) = (1, n, n-1)$$

Thus $rs = sr^{n-1}$ □

Definition 1.2.3. The dihedral group D_n is the symmetry group of a regular n-gon.

$$D_n = \langle s, r \mid r^n = s^2 = e, rs = sr^{n-1} \rangle$$

Claim: For all $1 \leq m \leq n-1$ we have $r^m s = sr^{n-m}$.

Proof. Base Case: $m = 1 \implies rs = sr^{n-1}$ ✓

Inductive Step: Assume for $k \geq 1$ the statement is true

Consider:

$$r^{k+1}s = rr^k s$$

$$= rsr^{n-k}$$

$$= sr^{n-1}r^{n-k}$$

$$= sr^n r^{n-(k+1)}$$

$$= sr^{n-(k+1)}$$

Apply inductive hypothesis on $r^k s$

□

1.2.4 Exercises

Example 1.2.4. Simplify

- $(sr^5)r(sr^3)$ in D_4

$$= (sr)r(rs)$$

$$= e$$

- $(sr^4)(sr^2)$ in D_6

$$= (r^2s)(sr^2) \quad \text{Use } r^m s = sr^{n-m}$$

$$= r^4$$

- $r^3(sr)$ in D_4

$$= r^3(r^3s)$$

$$= r^2s$$

Example 1.2.5. The Cayley table of the D_3 can be written

	e	r	r^2	s	rs	r^2s
e	e	r	r^2	s	rs	r^2s
r	r	r^2	e	rs	r^2s	s
r^2	r^2	e	r	r^2s	s	rs
s	s	r^2s	rs	e	r^2	r
rs	rs	s	r^2s	r	e	r^2
r^2s	r^2s	rs	s	r^2	r	e

1.3 Symmetric Groups

Definition 1.3.1. Given a set X , a permutation of X is a bijection $X \rightarrow X$
 $S_X = \{ \sigma : X \rightarrow X \mid \sigma \text{ is a bijection} \}$

Lemma 1.3.2. For all sets $X \neq \emptyset$, S_X is a group

Proof.

- Closure: Suppose $\sigma, \mu \in S_X$
 Consider $\sigma \circ \mu$ since it is a composition of bijections it is also a bijection.
 Also functions go from $X \rightarrow X$ thus $\sigma \circ \mu$ is the same.
 Therefore, the composition is a bijection that is mapped from $X \rightarrow X$ thus $\sigma \circ \mu \in S_X$
- Identity: Consider $e : X \rightarrow X$ s.t. $e(x) \mapsto x$
 Thus $e \in S_X$ and for all $\sigma \in S_X$ $\sigma \circ e(x) = \sigma(x)$
 Thus $\sigma \circ e = \sigma$. Also, it follows that $e \circ \sigma = \sigma$
- Inverses: Since S_X has only bijective functions all bijective functions have inverses.
- Associativity: We have associativity by the fact that function composition has associativity.

□

Definition 1.3.3. The permutation group on n -objects S_n is the S_X with $X = [n]$
 Observe $|S_n| = n!$

Example 1.3.4. $\sigma, \mu \in S_4$

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \mu = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix}$$

$$\sigma^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \mu^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix}$$

$$\sigma\mu = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \mu\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix}$$

1.3.1 Cycle Notation

Example 1.3.5. $\sigma, \mu \in S_5$

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 4 & 2 & 5 & 1 \end{pmatrix} \mu = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 5 & 2 & 1 & 3 \end{pmatrix}$$

$$\sigma = (13245) \mu = (14)(253)$$

Example Calculations:

$$\sigma\mu = (13245)(14)(253)$$

$$= (152)(34)$$

$$\mu\sigma = (14)(253)(13245)$$

$$= (12)(354)$$

$$\mu^2 = (14)(253)(14)(253)$$

$$= (14)^2(253)^2 = (235)$$

Lemma 1.3.6. *Disjoint cycles are commutative*

Proof. Let α, β be disjoint cycles of $[n]$

Let $\alpha = (a_1, a_2, \dots, a_k)$, and $\beta = (b_1, b_2, \dots, b_j)$

Let $x \in [n]$

Consider: $\alpha\beta(x), \beta\alpha(x)$

If $x \in \alpha$ then $x \notin \beta \implies \beta(x) = x \implies \alpha\beta(x) = \alpha(x) = a_i$
 $\implies \beta\alpha(x) = \beta(a_i) = a_i$ since $a_i \notin \beta$

Use similar reasoning for if $x \in \beta$

$\implies \alpha\beta = \beta\alpha$

□

Lemma 1.3.7. *Every $\sigma \in S_n$ can be written as disjoint cycles*

Lemma 1.3.8. *The order of an m -cycle is m*

Lemma 1.3.9. Suppose G is a group and $\text{ord}(g) = n$. Then $g^m = e \iff n \mid m$

Proof. Let G be a group and $\text{ord}(g) = n$

($\Leftarrow$)

Assume $n \mid m$

$$\implies m = nk, k \in \mathbb{Z}$$

$$g^m = g^{nk} = (g^n)^k = (e)^k = e$$

($\Rightarrow$)

Assume $g^m = e$ and use the division algorithm (0.1.1) to write $m = nq + r$ s.t. $0 \leq r < n$

$$e = g^m = g^{nq+r} = (g^n)^q g^r = g^r \implies r = 0 \implies n \mid m$$

□

Theorem 1.3.10. If $\sigma_1, \sigma_2, \dots, \sigma_k$ are disjoint cycles of length $a_1, a_2, \dots, a_k$ and $\sigma = \sigma_1 \sigma_2 \dots \sigma_k$. Then $\text{ord}(\sigma) = \text{lcm}(a_1, a_2, \dots, a_k)$.

Proof. Let $m = \text{ord}(\sigma)$, $l = \text{lcm}(a_1, a_2, \dots, a_k)$

Since l is a lcm it is also a multiple so $l = a_j q_j$ for $1 \leq j \leq k$ where $q_j \in \mathbb{Z}$

$$\sigma^l = \sigma_1^l \sigma_2^l \dots \sigma_k^l$$

We can do this because of Lemma 1.3.6

$$= (\sigma_1^{a_1})^{q_1} \dots (\sigma_k^{a_k})^{q_k}$$

$$= (1)$$

Apply Lemma 1.3.8

$$\implies m \mid l$$

Also, $(1) = \sigma^m = \sigma_1^m \dots \sigma_k^m$ since they are disjoint none of them are inverses of each other

$$\implies \sigma_1^m = \dots = \sigma_k^m = (1)$$

$$(\text{Lemma 2.1.9}) \implies a_1 \mid m \dots a_k \mid m \implies l \mid m$$

Thus, since $l \mid m$ and $m \mid l$ $m = l$

□

Example 1.3.11.

- Find an element of order 12 in S_7
 $(1234)(567) \in S_7$ by Theorem 1.3.10 $\implies$ the order is $3 \cdot 4 = 12 = \text{lcm}(3,4)$
- Possible orders of elements in S_5
 - 1: (1) , $5=1+1+1+1+1$
 - 2: (12) , $5=2+1+1+1+1$ or $(12)(34)$, $5=2+2+1$
 - 3: (123) , $5=3+1+1$
 - 4: (1234) , $5=4+1$
 - 5: (12345) , $5=5$
 - 6: $(12)(345)$, $5=2+3$

1.3.2 Exercises

Example 1.3.12. Write the following as the product of disjoint cycles and find the order.

- $(1358)(378)(125) \in S_8$
 (128537) order=6
- $(1264)^{-1}(364)^2 \in S_6$
 $= (1462)(346)$
 $= (142)(36)$ order=6

Example 1.3.13. Find all Possible orders of elements in S_7

- 1: (1) , $7=1+1+1+1+1+1+1$
- 2: (12) , $7=2+1+1+1+1+1$ or $(12)(34)$, $7=2+2+1+1+1$ or $(12)(34)(56)$, $7=2+2+2+1$
- 3: (123) , $7=3+1+1+1+1$ or $(123)(456)$, $7=3+3+1$
- 4: (1234) , $7=4+1+1+1$ or $(1234)(56)$, $7=4+2+1+1$
- 5: (12345) , $7=5+1+1$
- 6: (123456) , $7=6+1$ or $(123)(45)$, $7=3+2+1+1$, or $(123)(45)(67)$, $7=3+2+2+1$
- 7: (1234567) , $7=7$
- 10: $(12345)(67)$, $7=5+2$
- 12: $(1234)(567)$, $7=4+3$

Example 1.3.14. What is the largest possible order of an element in S_{10} ?

cycle type	order
(5,1,1,1,1)	5
(5,2,1,1,1)	10
(5,2,2,1)	10
(5,3,1,1)	15
(5,3,2)	30
(5,4,1)	20
(5,5)	5
(6,1,1,1,1)	6
(6,2,1,1)	6
(6,3,1)	6
(6,4)	12
(7,1,1,1)	7
(7,2,1)	14
(7,3)	21
(8,1,1)	8
(8,2)	8
(9,1)	9
(10)	10

Max=30

1.4 General Results on Groups

Lemma 1.4.1. *Suppose G is a group. Then*

- (1) *The identity is unique*
- (2) *Inverses are unique*
- (3) *For every $a, b, c \in G$ there is a unique $x \in G$ s.t. $ax = b$*
- (4) *If $ab = ac$ then $b = c$, similarly if $xy = zy$ then $x = z$*
- (5) $(ab)^{-1} = b^{-1}a^{-1}$
- (6) *If $(ab)^2 = a^2b^2$ then $ab = ba$*
- (7) *If $\text{ord}(g) = 2$ for all $g \in G$ then G is abelian*

• *Proof.* (3) Let $ax = b \implies x = a^{-1}b$

Suppose $x, y \in G$ s.t. $ax = b$ and $ay = b$

$$ax = ay \implies x = y \quad \square$$

• *Proof.* (6) Assume $(ab)^2 = a^2b^2$

$$ab = (a^{-1}a)(ab)(bb^{-1}) = a^{-1}a^2b^2b^{-1} = a^{-1}(ab)^2b^{-1} = (a^{-1}a)(ba)(bb^{-1}) = ba \quad \square$$

• *Proof.* Assume $\text{ord}(g) = 2$ for all $g \in G$

$$\implies g = g^{-1} \text{ for all } g \in G$$

Let $a, b \in G$. Consider ab

$$ab = (ab)^{-1} = b^{-1}a^{-1} = ba \quad \square$$

Definition 1.4.2. Given a group G and a subset $H \subseteq G$, we say H is a subgroup $H \leq G$ if

- The identity of G is in H
- For all $h \in H, h^{-1} \in H$
- For all $a, b \in H, ab \in H$

Theorem 1.4.3. Suppose that G is a group and $\emptyset \neq H \subseteq G$
 then $H \leq G \iff$ for all $a, b \in H, ab^{-1} \in H$

Proof.

($\implies$) Assume $H \leq G$

Let $a, b \in H$

By Definition 1.4.2 $\implies b^{-1} \in H \implies ab^{-1} \in H$

($\impliedby$) Assume for all $a, b \in H, ab^{-1} \in H$

- (Identity) Consider $a \in H$

$$\implies aa^{-1} = e \in H$$

- (Inverses) $\checkmark$

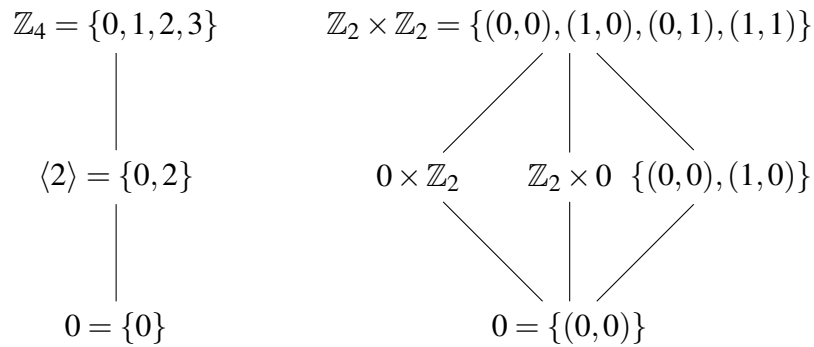
- (Closure) Consider $a, b^{-1} \in H$

$$\implies ab \in H$$

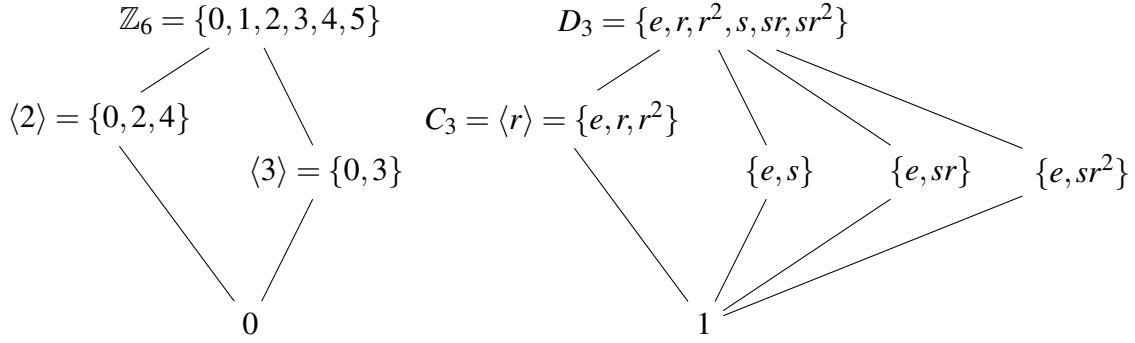
□

1.4.1 Subgroup Lattice

Subgroup of Order 4



Subgroup of Order 6



Definition 1.4.4. Given a group G and a subset $S \subseteq G$

- The centralizer of S in G : $C_G(S) = \{g \in G \mid gs = sg, \forall s \in S\}$
- The center of G : $(C_G(G)) = Z(G) = \{z \in G \mid gz = zg, \forall g \in G\}$

Lemma 1.4.5. $C_G(S)$ and $Z(G)$ are subgroups of G

Proof. Both contain e so they are nonempty

Let $a, b \in C_G(S)$ and $s \in S$

Consider:

$$\begin{aligned}
 as &= sa \\
 sab^{-1} &= asb^{-1} \\
 &= a(b^{-1}s)b \\
 &= (ab^{-1})s
 \end{aligned}$$

Thus $ab^{-1} \in C_G(S)$

Thus, by Theorem 1.4.3 $C_G(S)$ is a subgroup

$\implies Z(G) \leq G$ since the center is a special case of the centralizer

□

Example 1.4.6. Centralizers in D_4

- $C_{D_4}(e) = D_4 = C_{D_4}(r^2)$ ²
- $C_{D_4}(r) = e, r, r^2, r^3 = C_{D_4}(r^3)$
- $C_{D_4}(s) = e, s, r^2, sr^2 = C_{D_4}(sr^2)$
- $C_{D_4}(sr) = e, sr, r^2, sr^3 = C_{D_4}(sr^3)$
- $Z(D_4) = e, r^2$

²Since $r^2s = sr^2$ and $r^2r = rr^2$

1.4.2 Exercises

Example 1.4.7. Prove that if G is abelian if and only if for all $a, b \in G$ we have $(ab)^{-1} = a^{-1}b^{-1}$

Proof. ($\implies$) Assume G is abelian and let $a, b \in G$

$$\implies (ab)^{-1} = b^{-1}a^{-1} = a^{-1}b^{-1}$$

($\impliedby$) Assume for all $a, b \in G$ we have $(ab)^{-1} = a^{-1}b^{-1}$

Consider: a^{-1} and b^{-1}

$$ba = (a^{-1}b^{-1})^{-1} = ab$$

□

Example 1.4.8. Suppose G is a group of order 6 containing 7 and 19 where the operation is multiplication modulo 36. List all the elements of G .

- $7 \times 7 = 49 = 13$
- $19 \times 19 = 361 = 1$
- $7 \times 19 = 133 = 25$
- $13 \times 19 = 247 = 31$

$$G = \{1, 7, 13, 19, 25, 31\}$$

Example 1.4.9. Suppose that G is a group such that for all $a, b, c, d \in G$ such that if

$$axb = cxd$$

then $ab = cd$. Show that G is abelian

Proof. Consider: $a, b \in G$

$$aa^{-1}b = ba^{-1}a \implies ab = ba$$

□

1.5 Cyclic Groups

Definition 1.5.1. Given a group G and an element $g \in G$, the cyclic subgroup generated by g is $\langle g \rangle = \{g^m \mid m \in \mathbb{Z}\} = \{\dots, g^{-2}, g^{-1}, e, g, g^2, \dots\}$

We say G is a cyclic group if there is a $g \in G$ s.t. $G = \langle g \rangle$

$$\text{ord}(g) = |\langle g \rangle|$$

Cyclic subgroups of $\mathbb{Z}$

Example 1.5.2.

- $\langle 2 \rangle = \{2m \mid m \in \mathbb{Z}\} = \{\dots, -4, -2, 0, 2, 4, \dots\} = 2\mathbb{Z}$
- $\langle n \rangle = \{nm \mid m \in \mathbb{Z}\} = \{\dots, -2n, -n, 0, n, 2n, \dots\} = n\mathbb{Z}$

Cyclic subgroups of $\mathbb{Q}^\times$

Example 1.5.3.

- $\langle -1 \rangle = \{-1, 1\}$
- $\langle 2 \rangle = \{2^m \mid m \in \mathbb{Z}\}$

Cyclic subgroups of $\mathbb{C}^\times$

Example 1.5.4.

$$\langle i \rangle = \{i, -1, -i, 1\}$$

Cyclic subgroups of $\mathbb{Z}_{12}$

Example 1.5.5.

- $\mathbb{Z}_{12} = \langle 1 \rangle = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$
- $\langle 2 \rangle = \{0, 2, 4, 6, 8, 10\}$
- $\langle 4 \rangle = \{0, 4, 8\}$
- $\langle 3 \rangle = \{0, 3, 6, 9\}$

Cyclic subgroups of $U_9 = \{1, 2, 4, 5, 7, 8\}$

Example 1.5.6.

- $\langle 1 \rangle = \{1\}$
- $\langle 8 \rangle = \{8, 64\} = \{1, 8\}$
- $\langle 4 \rangle = \{4, 16, 64\} = \{1, 4, 7\}$
- $\langle 2 \rangle = \{2, 4, 8, 16, 32, 64\} = \{1, 2, 4, 5, 7, 8\} = U_9$

Cyclic subgroups of D_n

Example 1.5.7.

- $\langle r \rangle = \{e, r, r^2, \dots, r^{n-1}\} = Z_n^3 = C_n^4$
- $\langle s \rangle = \{e, s\}$
- $\langle sr \rangle = \{e, sr\}$

Cyclic subgroups of S_5

Example 1.5.8.

$$\langle (12)(345) \rangle^5 = \{(1), (12)(345), (354), (12), (345), (12)(354)\}$$

Theorem 1.5.9. Suppose $g \in G$ with $\text{ord}(g) = n$ and let $m \in \mathbb{N}$ then, $\langle g^m \rangle = \langle g^{\gcd(m,n)} \rangle$

Proof. Set $d = \gcd(m, n)$.

Take $x \in \langle g^m \rangle \implies x = (g^m)^k = g^{mk}$ for some $k \in \mathbb{Z}$

Also, $m = dq$ for some $q \in \mathbb{Z}$ since d is a divisor of m

$x = g^{dqk} = (g^d)^{qk} \in \langle g^d \rangle$ so $\langle g \rangle \subseteq \langle g^d \rangle$

By Lemma 0.1.2 there exists $r, s \in \mathbb{Z}$ s.t. $mr + ns = d$

Take $x \in \langle g^d \rangle \implies x = g^{dk}$ for some $k \in \mathbb{Z}$

$g^{dk} = g^{k(mr+ns)} = g^{mkr}(g^n)^{ks} = g^{mkr} \in \langle g^m \rangle$ so $\langle g^d \rangle \subseteq \langle g^m \rangle$

Thus $\langle g^m \rangle = \langle g^d \rangle$

□

³not $\mathbb{Z}$

⁴Multiplicative cyclic group of order n

⁵By Theorem 1.3.10 the $\text{lcm}(2,3)=6=\text{order}$ and thus this cyclic group will have 6 elements

Theorem 1.5.10 (Fundamental Theorem of Cyclic Groups).

- I. Every subgroup of a cyclic group is cyclic
- II. If G is a cyclic group of order n and $d \mid n$ then there is a unique subgroup $H \leq G$ with $|H| = d$

I. *Proof.* Suppose $G = \langle g \rangle$ and $H \leq G$

Case 1: $H = \{e\} = \langle e \rangle$ ✓

Case 2: $H \neq \{e\}$. Thus, $H = \{\dots, g^{-m_2}, g^{-m_1}, e, g^{m_1}, g^{m_2}, \dots\}$ s.t. $m_i \in \mathbb{N}$

By the Well Ordering Principle, there exists a minimum m_i call this m

Since $g^m \in H \implies (g^m)^n \in H \ \forall n \in \mathbb{Z} \implies \langle g^m \rangle \subseteq H$

Take $h \in H$, write $h = g^k$ for $k \in \mathbb{Z}$

By the division algorithm (Thm. 0.1.1), there exists a $q, r \in \mathbb{Z}$ s.t.

$k = mq + r$ with $0 \leq r < m$

$g^r = (g^k)(g^{-mq})$ thus since $g^k \in H$ and $g^{-mq} \in \langle g^m \rangle \subseteq H \implies g^r \in H$

Since $0 \leq r < m$ and m is the minimum $\implies r = 0$

Thus $h = (g^m)^q \in \langle g^m \rangle \implies H \subseteq \langle g^m \rangle$

$\implies H = \langle g^m \rangle \implies H$ is cyclic □

II. *Proof.* Assume G is a cyclic group of order n and $d \mid n$

Let $G = \langle g \rangle$

Existence: Consider $\langle g^{\frac{n}{d}} \rangle$ this has d elements so ✓

Uniqueness: Let $H \leq G$, $|H| = d$ by Theorem 1.5.10.I

$\implies H = \langle g^m \rangle$ by Theorem 1.5.9 $\implies H = \langle g^{\gcd(m,n)} \rangle$

Thus $|H| = d = \frac{n}{\gcd(m,n)} \implies \gcd(m,n) = \frac{n}{d} \implies H = \langle g^{\frac{n}{d}} \rangle$ which unique describes a group □

Definition 1.5.11 (Euler's Totient Function).

For $n \in \mathbb{N}$, define $\varphi(n) = \left| \{1 \leq m \leq n \mid \gcd(m,n) = 1\} \right|$

Theorem 1.5.12.

Suppose G is a cyclic group of order n and $d \mid n$ then the number of elements of G of order d is $\varphi(d)$

Proof. Assume G is cyclic group of order n and $d \mid n$

Let $H \leq G$ be the unique subgroup s.t. $|H| = d \iff$ Theorem 1.5.10.II

Let $H = \langle h \rangle \iff$ Theorem 1.5.10.I

Let $x \in G$ have order d

Consider: $\langle x \rangle$ since x is order $d \implies |\langle x \rangle| = d$

Since H is unique $\implies \langle h \rangle = \langle x \rangle \implies x = h^m$ where $\gcd(m, d) = 1 \iff$ Theorem 1.5.9

The number of choices of m is $\varphi(d)$

□

1.5.1 Exercises

Example 1.5.13. Find all generators of $\mathbb{Z}_{15}$ and $\mathbb{Z}_{12}$

By Theorem 1.5.9 this implies all the numbers that are relatively prime to 12 and 15

Thus, the generators for $\mathbb{Z}_{15}$ are 1,2,4,7,8,11,13,14

Thus, the generators for $\mathbb{Z}_{12}$ are 1,5,7,11

Example 1.5.14. Show that $\mathbb{Q}_{\geq 0}^{\times}$ is not cyclic

Assume $\mathbb{Q}_{\geq 0}^{\times}$ is cyclic thus there exists a $q = \frac{a}{b} \in \mathbb{Q}_{\geq 0}^{\times}$ s.t. $\mathbb{Q}_{\geq 0}^{\times} = \langle q \rangle$

Consider: p a prime bigger than a, b

Let $\frac{1}{p} = \frac{a^m}{b^m}$

Case 1: $m \geq 1$

$\implies b^m = pa^m \implies p \mid b^m \implies \Leftarrow$ since we picked p to be bigger than a, b

Case 2: $m \leq -1$

$\implies a^m = pb^m \implies p \mid a^m \implies \Leftarrow$

1.6 Homomorphism and Isomorphism

Definition 1.6.1. Let $(G_1, *)$ and $(G_2, \circ)$ be groups. A map $\varphi : G_1 \rightarrow G_2$ is called a homomorphism if for all $a, b, c \in G_1$ we have $\varphi(a * b) = \varphi(a) \circ \varphi(b)$

If, in addition, φ is a bijection we say φ is an isomorphism and G_1 and G_2 are isomorphic

$$G_1 \cong G_2$$

Theorem 1.6.2. Suppose $G = \langle g \rangle$ is a cyclic group.

If $|G| = \infty$ then $G \cong \mathbb{Z}$

If $|G| = n$ then $G \cong \mathbb{Z}_n$

Proof. Assume $|G| = \infty$ Define:

$$\begin{aligned} \varphi : G &\rightarrow \mathbb{Z} \\ g^m &\mapsto m \end{aligned}$$

$$\begin{aligned} \varphi^{-1} : \mathbb{Z} &\rightarrow G \\ m &\mapsto g^m \end{aligned}$$

Let $g^a, g^b \in G$ and $c, d \in \mathbb{Z}$

Consider: $\varphi(g^a g^b) = \varphi(g^{a+b}) = a + b = \varphi(g^a) + \varphi(g^b)$

Consider: $\varphi^{-1}(c + d) = g^{c+d} = g^c g^d = \varphi^{-1}(c) + \varphi^{-1}(d)$

Consider: $\varphi \circ \varphi^{-1}(c) = \varphi(g^c) = c$

$\varphi^{-1} \circ \varphi(g^a) = \varphi^{-1}(a) = g^a$

Thus, φ is a homomorphism and bijective

$$\implies G \cong \mathbb{Z}$$

Assume $|G| = n$

$$\begin{aligned} \varphi : G &\rightarrow \mathbb{Z}_n \\ g^m &\mapsto [m] \end{aligned}$$

$$\begin{aligned} \varphi^{-1} : \mathbb{Z}_n &\rightarrow G \\ [m] &\mapsto g^m \end{aligned}$$

homomorphism: $\checkmark$

Consider: $\varphi \circ \varphi^{-1}([c]) = \varphi(g^c) = [c]$

$\varphi^{-1} \circ \varphi(g^a) = \varphi^{-1}([a]) = g^a$

Thus, φ is a homomorphism and bijective

$\implies G \cong \mathbb{Z}_n$

□

Theorem 1.6.3 (Cayley's Theorem). *Every group is isomorphic to a permutation group.*

Proof. Let G be a group and by Lemma 1.3.2 $\implies S_G$ is a group

Define:

$$\varphi : G \rightarrow S_G$$

$$g \mapsto l_g$$

$$l_g : G \rightarrow G$$

$$x \mapsto gx$$

- **Injective:** Let $x, y \in G$ s.t. $l_g(x) = l_g(y)$

$$\implies gx = gy \implies x = y$$

- **Surjective:** Let $y \in G$

$$\text{Consider: } l_g(g^{-1}y) = gg^{-1}y = y$$

Thus $l_g \in S_G$

Consider: $a, b \in G$

$$\varphi(ab)(x) = l_{ab}(x) = abx = a(bx) = l_a \circ l_b(x) = \varphi(a)\varphi(b)(x)$$

Injective: $a, b, x \in G$ s.t. $\varphi(a) = \varphi(b)$

$$l_a = l_b \implies l_a(x) = l_b(x) \quad \forall x \in G \implies ax = bx \implies ae = be \implies a = b$$

$$\implies G \cong \text{im}(G) \leq S_G$$

□

Theorem 1.6.4. Suppose $\varphi : G_1 \rightarrow G_2$ is a homomorphism then

I. $\varphi(e_1) = e_2$

II. $\forall n \in \mathbb{Z}, \varphi(g^n) = [\varphi(g)]^n$

III. $ab = ba \implies \varphi(a)\varphi(b) = \varphi(b)\varphi(a)$

IV. $\text{ord}(\varphi(g)) \mid \text{ord}(g)$

V. $H \leq G_1 \implies \varphi(H) \leq G_2$

VI. $K \leq G_2 \implies \varphi^{-1}(K) \leq G_1$

I. *Proof.* $e_2\varphi(e_1) = e_2\varphi(e_1e_1) = e_2\varphi(e_1)\varphi(e_1) \implies e_2 = \varphi(e_1)$ □

II. *Proof.* Base Case: $n = 0$ $\varphi(g^0) = \varphi(e_1) = e_2 = [\varphi(g)]^0$

Assume statement is true for $k \geq 0$ for $k \in \mathbb{N}$

Consider: $k + 1$

$$\varphi(g^{k+1}) = \varphi(g)\varphi(g^k) = \varphi(g)[\varphi(g)]^k = [\varphi(g)]^{k+1}$$

Note: $e_2 = \varphi(e_1) = \varphi(gg^{-1}) = \varphi(g)\varphi(g^{-1}) \implies [\varphi(g)]^{-1} = \varphi(g^{-1})$

This proves the statement for all negative integers since we just apply the induction on g then find its inverse □

III. *Proof.* $\varphi(a)\varphi(b) = \varphi(ab) = \varphi(ba) = \varphi(b)\varphi(a)$ □

IV. *Proof.* Set $n = \text{ord}(g)$

$$\implies \varphi(g)^n = \varphi(g^n) = \varphi(e_1) = e_2 \implies \text{ord}(\varphi(g)) \mid \text{ord}(g)$$
 □

V. *Proof.* Let $x, y \in \varphi(H)$ s.t. $\varphi(a) = x$ and $\varphi(b) = y$ for $a, b \in H$

$$\implies ab^{-1} \in H \implies \varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} \in \varphi(H)$$

Thus, by Theorem 1.4.3 $\varphi(H) \leq G_2$ □

VI. *Proof.* Let $a, b \in \varphi^{-1}(K) \implies \varphi(a), \varphi(b) \in K$

$$\implies \varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} \in K \implies ab^{-1} \in \varphi^{-1}(K)$$

Thus, by Theorem 1.4.3 $\varphi^{-1}(K) \leq G_1$ □

Theorem 1.6.5. *thm:isomorphismgeneral Suppose $\varphi : G_1 \rightarrow G_2$ is an isomorphism then φ^{-1} is an isomorphism*

II. G_1 is abelian $\iff G_2$ is abelian

III. G_1 is cyclic $\iff G_2$ is cyclic

IV. $\text{ord}(g) = \text{ord}(\varphi(g))$

I. $\checkmark$

II. *Proof.* Suppose G_1 is abelian and $x, y \in G_2$

$$xy = \varphi(a)\varphi(b) = \varphi(ab) = \varphi(ba) = \varphi(b)\varphi(a) = yx$$

Do same thing for other direction $\square$

III. *Proof.* Suppose G_1 is cyclic and $\langle g \rangle = G_1$

$$\text{Take } x \in G_2 \implies \varphi(a) = x$$

$$a = g^n \implies x = \varphi(g^n) = \varphi(g)^n \implies x \in \langle \varphi(g) \rangle \implies G_2 \subseteq \langle \varphi(g) \rangle$$

$$\text{Take } x \in \langle \varphi(g) \rangle \implies x = \varphi(g)^n$$

$$\implies x = \varphi(g^n) \implies x \in G_2 \implies \langle \varphi(g) \rangle \subseteq G_2$$

$$\implies G_2 = \langle \varphi(g) \rangle \implies G_2 \text{ is cyclic}$$

Do same thing for other direction $\square$

IV. *Proof.* By Theorem 1.6.4.IV $\text{ord}(g) \mid \text{ord}(\varphi(g))$ and $\text{ord}(\varphi(g)) \mid \text{ord}(g) \implies \text{ord}(g) = \text{ord}(\varphi(g))$

$\square$

1.6.1 Exercises

Example 1.6.6. Describe all homomorphisms $\varphi : \mathbb{Z}_{12} \rightarrow \mathbb{Z}_{20}$

$$\text{ord}(1) = 12 \text{ in } \mathbb{Z}_{12} \implies \text{ord}(\varphi(1)) \mid 12 \implies \text{ord}(\varphi(1)) \in \{1, 2, 3, 4, 6, 12\}$$

$$\text{ord}(m) = \frac{n}{\gcd(m,n)} \text{ in } \mathbb{Z}_n$$

$$\bullet \text{ ord}(\varphi(1)) = 1 \implies \varphi(1) = 0 \implies \varphi(n) = 0$$

$$\bullet \text{ ord}(\varphi(1)) = 2 \implies \varphi(1) = 10$$

$$\text{this is the only element with a gcd of 10 with 20} \implies \varphi(n) = 10n$$

$$\bullet \text{ ord}(\varphi(1)) = 3, 6, 12 \text{ there is no element with these orders since these are not divisors of 10}$$

$$\bullet \text{ ord}(\varphi(1)) = 4 \implies \varphi(1) = 5, 15 \implies \varphi(n) = 5n, 15n$$

Example 1.6.7. Show that $\mathbb{R} \not\cong \mathbb{R}^\times$

$$\text{Assume } \mathbb{R} \cong \mathbb{R}^\times \text{ via } \varphi : \mathbb{R}^\times \rightarrow \mathbb{R}$$

$$\text{Take } x \in \mathbb{R}^\times \text{ s.t. } \varphi(x) = -1$$

$$\text{Consider } \varphi\left(\frac{x}{2}\right)^2 = \varphi(x) = -1 \implies \varphi\left(\frac{x}{2}\right) \in \mathbb{R} \text{ with } \varphi\left(\frac{x}{2}\right)^2 = -1 \Rightarrow \Leftarrow$$

1.7 Cosets

Definition 1.7.1. Suppose G is a group $H \leq G$, and $g \in G$.

- The left coset of H with representative g is

$$gH = \{gh \mid h \in H\}$$

- The right coset of H with representative g is

$$Hg = \{hg \mid h \in H\}$$

Example 1.7.2. $G = S_3, H = \{(1), (12)\}$

$$(13)H = \{(13), (123)\} = (123)H$$

$$(23)H = \{(23), (132)\} = (132)H$$

$$H(13) = \{(13), (132)\} = H(132)$$

$$H(23) = \{(23), (123)\} = H(123)$$

Theorem 1.7.3. Suppose that G is a group, $H \leq G$, $x, y \in G$. The following are equivalent

I. $xH = yH$

II. $Hx^{-1} = Hy^{-1}$

III. $xH \subseteq yH$

IV. $y \in xH$

V. $x^{-1}y \in H$

Proof. (I. $\implies$ II.)

Assume I. Let $g \in Hx^{-1} \implies g = hx^{-1}$ for $h \in H \implies g^{-1} = xh^{-1} \in xH = yH$

$\implies g^{-1} = yj$ for $j \in H \implies g = j^{-1}y^{-1} \in Hy^{-1} \implies Hx^{-1} \subseteq Hy^{-1}$

Do the same thing for other direction thus $Hx^{-1} = Hy^{-1}$

(II. $\implies$ I.)

Assume II. Let $g \in xH \implies g = xh$ for $h \in H \implies g^{-1} = h^{-1}x^{-1} \in Hx^{-1} = Hy^{-1}$

$\implies g^{-1} = jy^{-1}$ for $j \in H \implies g = yj \in Hy \implies xH$

(III. $\implies$ IV.)

Assume III. Let $g \in xH \subseteq yH \implies g = xh = yj$ for $h, j \in H \implies y = xhj^{-1} \implies y \in xH$

(IV. $\implies$ V.)

Assume IV. $y \in xH \implies y = xh$ for $h \in H \implies x^{-1}y = h \in H$

($V \implies I.$)

Assume V. Let $g \in xH \implies g = xh$ for $h \in H \implies g = y(y^{-1}xh) \in yH$ since $y^{-1}x \in H$

Do the same thing for the other direction this $xH = yH$ □

Theorem 1.7.4. Suppose $H \leq G$. The cosets of H partition G . that is, they are disjoint and they union to G .

Proof.

Disjoint: Suppose $x, y \in G$ s.t. $xH \cap yH \neq \emptyset$

Take $g \in xH \cap yH \implies g = xh = yj$ $h, j \in H \implies x^{-1}y = hj^{-1} \in H$

Thus, by Theorem 1.7.3 $xH = yH$

Union: $\bigcup_{x \in G} xH \subseteq G \iff xH \subseteq G$

Take $g \in G$ by Theorem 1.7.3 $\implies g \in gH \implies g \in \bigcup_{x \in G} xH \implies G \subseteq \bigcup_{x \in G} xH$

$\implies G = \bigcup_{x \in G} xH$ □

Definition 1.7.5. Let G be any group with $H \leq G$.

Define the index of H in G to be the number of left(right) cosets of H in G , denote

$$[G : H]$$

Theorem 1.7.6. The number of left cosets is equal to the number of right cosets

Proof. Let $\mathfrak{L}_H = \{gH \mid g \in G\}$ and $\mathfrak{R}_H = \{Hg \mid g \in G\}$

Define:

$$f: \mathfrak{L}_H \rightarrow \mathfrak{R}_H$$

$$gH \mapsto Hg^{-1}$$

Well Defined: Suppose $xH = yH$, $f(xH) = Hx^{-1} = Hy^{-1} = f(yH)$

Injective: Suppose $f(xH) = f(yH) \implies Hx^{-1} = Hy^{-1} \implies xH = yH$

Surjective: Let $Hg \in \mathfrak{R}_H$, $f(g^{-1}H) = Hg$

Therefore, $|\mathfrak{L}_H| = |\mathfrak{R}_H|$ □

1.7.1 Exercices

Example 1.7.7. Find all left and right cosets of $\langle s, r \rangle \leq D_6$

1. $eH = He = H$

2. $rH = \{r, r^3, r^5, sr, sr^3, sr^5\} = Hr$

Since these union to the full set we are done

Example 1.7.8. Suppose that $[G : H] = 2$ and $a, b \notin H$. Show that $ab \in H$

Since $a \notin H \implies a^{-1} \notin H$

Since $a^{-1}, b \notin H \implies a^{-1}H \neq H$ and $bH \neq H$

$[G : H] = 2 \implies a^{-1}H = bH \implies$ by Theorem 1.7.3. $\forall ab \in H$

1.8 Lagrange's Theorem

Theorem 1.8.1 (Lagrange's Theorem). *If G is a finite group with $H \leq G$ then $|G| = [G : H]|H|$. In particular the order of a subgroup divides the order of a finite group*

Proof. Set $m = [G : H]$ and take $g_1, g_2, \dots, g_m \in G$ so that $g_1H, g_2H, \dots, g_mH$ are all distinct.

Also, $|g_iH| = |H| \ \forall 1 \leq i \leq m$. This is because you take all elements of H and add g_i to them.

Since $G = g_1H \cup g_2H \cup \dots \cup g_mH$ and these are all disjoint by Theorem 1.7.4

By the multiplication principle, $|G| = m|H| = [G : H]|H|$ □

Corollary 1.8.2. *If $|G| < \infty$ and $g \in G$ then $\text{ord}(g) \mid |G|$ thus $g^{|G|} = e$*

Proof. Take $H = \langle g \rangle$ and $\text{ord}(H) = \text{ord}(g)$ thus by Theorem 1.8.1 $\text{ord}(g) = \text{ord}(H) \mid |G|$

Thus by Lemma 1.3.9 $g^{|G|} = e$ □

Example 1.8.3. Suppose $|G| = 48$ and $K \leq H \leq G$ with $|K| = 8$.

What are the possible values of $|H|$.

By Theorem 1.8.1 we know $|H| \mid |G| = 48$ and $|K| = 8 \mid |H|$

Thus $|H| = 8, 16, 24, 48$

Example 1.8.4 (The Converse of Lagrange's Theorem is false).

Corollary 1.8.5. *If p is prime and $|G| = p$ then $G \cong \mathbb{Z}_p$*

Proof. Take $e \neq g \in G$ and note $\langle g \rangle \leq G$ also $|\langle g \rangle| \neq 1$

but by Theorem 1.8.1 $|\langle g \rangle| \mid p$ since p is prime $\implies |\langle g \rangle| = p \implies \langle g \rangle = G$

By Theorem 1.6.2 $G \cong \mathbb{Z}_p$ □

Corollary 1.8.6. *For all $a \in \mathbb{Z}, n \in \mathbb{N}$ s.t. $\gcd(a, n) = 1$ we have $a^{\varphi(n)} \equiv 1 \pmod{n}$*

Proof. Note $[a] \in U_n$, also $|U_n| = \varphi(n)$

$[a] \mid |U_n| = \varphi(n) \implies [a]^{\varphi(n)} = 1 \in U_n$

$\implies [a^{\varphi(n)}] = [1] \in U_n \implies a^{\varphi(n)} \equiv 1 \pmod{n}$ □

Theorem 1.8.7. *Let p be an odd prime and G a group of order $2p$ then $G \cong \mathbb{Z}_{2p}$ or $G \cong D_p$*

Proof.

Case 1: We have $g \in G$ s.t. $\text{ord}(g) = 2p \implies G = \langle g \rangle \cong \mathbb{Z}_{2p}$

Case 2: G does not have elements of order $2p$.

Assume G does not have an element of order $p \implies$ all non identity elements have order 2.

Take $e \neq a \neq b \neq e \in G$

Consider: $\{e, a, b, ab\} \leq G \implies 4 \mid 2p \implies 2 \mid p \Rightarrow \Leftarrow$

Thus, there exists a $h \in G$ s.t. $\text{ord}(h) = p$ and form $H = \langle h \rangle \leq G$

$[G : H] = \frac{|G|}{|H|} = \frac{2p}{p} = 2$. Take $g \notin H \implies G = H \sqcup gH$

Consider: g^2H

Case 2a: $g^2H = H \implies g^2 \in H$

$\implies \text{ord}(g^2) \mid p \implies \text{ord}(g^2) = 1$ thus $g^2 = e$ or $\text{ord}(g^2) = p$ thus $g^{2p} = e$

$\implies \text{ord}(g) = p$ since no element has order $2p \implies \langle g \rangle = \langle g^2 \rangle \leq H \implies g \in H \Rightarrow \Leftarrow$

$\implies \text{ord}(g) = 2$

Case 2b: $g^2H = gH \implies g^{-1}g^2 = g \in H \Rightarrow \Leftarrow$

Thus, anything not in H has order 2

$hg \in Hg \implies hg \notin H \implies \text{ord}(hg) = 2$

$e = (hg)^2 = hghg \implies g^{-1}h^{-1} = hg \implies gh^{p-1} = hg \implies hg^k = gh^{p-k}$

Cosets are a partition so

$G = \{e, h, h^2, \dots, h^{p-1}, g, gh, \dots, gh^{p-1}\} = \langle h, g \mid e = h^p = g^2, hg = gh^{p-1} \rangle = D_p$

□

1.8.1 Exercises

Example 1.8.8. Let G be a group of order pqr where p, q and r are distinct primes. If $H, K \leq G$ s.t. $|H| = pq$ and $|K| = qr$, show that $|H \cap K| = q$

Since $H \cap K \leq H$ and $H \cap K \leq K$

$$|H \cap K| \mid |H| = qp \text{ and } |H \cap K| \mid |K| = qr$$

$$\implies |H \cap K| = 1 \text{ or } q$$

Case 1: $|H \cap K| = q$

Case 2: $|H \cap K| = 1$

Consider: $HK = \{hk \mid h \in H, k \in K\}$

$|HK| = |H \times K|$ however we are over counting here if hk is the same element as another $h'k'$

$$h_1k_1 = h_2k_2 \implies h_2^{-1}h_1 = k_2k_1^{-1} \in H \cap K \text{ thus we over count by } |H \cap K|$$

$$\text{Thus } |HK| = \frac{|H||K|}{|H \cap K|} = pq^2r > pqr \Rightarrow \Leftarrow$$

$$\text{Thus } |H \cap K| = q$$

1.9 Normal Subgroups and Quotient Groups

Definition 1.9.1. Given a group G , we say N is a normal subgroup and write $N \trianglelefteq G$ if for all $g \in G, n \in N$ we have $gng^{-1} \in N \iff gN = Ng \iff gNg^{-1} = N$

In this case we can form the quotient group $G/N = \{gN \mid g \in G\}$ with $(xN)(yN) = (xy)N$

Theorem 1.9.2. Given a group G : $N \trianglelefteq G \iff G/N$ is a group

Proof. ($\Leftarrow$) Assume G/N is a group

Let $a, b, c, d \in G$ s.t. $aN = cN$ and $bN = dN$ then $abN = cdN$

$\implies ca^{-1} = j, bd^{-1} = h, abd^{-1}c^{-1} \in N$

$ahc^{-1} = ah(ja)^{-1} = aha^{-1}j^{-1} \in N \implies aha^{-1} \in N \implies N \trianglelefteq G$

($\implies$) Assume $N \trianglelefteq G$

Let $a, b, c, d \in G$ s.t. $aN = cN$ and $bN = dN \implies a \in bN \implies a = bn$ same for $c = dh$ and $dn = jd$ since $dN = Nd$

$$\begin{aligned} (aN)(cN) &= (ac)N \\ &= (bndh)N \\ &= (bdjhn)N \\ &= (bd)N \\ &= (bN)(dN) \end{aligned}$$

□

1.9.1 Exercises

Example 1.9.3. Show if $[G : H] = 2$ then $H \trianglelefteq G$. What is the isom. class of G/H ?

Assume $[G : H] = 2 \implies gH \sqcup H = G$ for $g \notin H$

Let $x \in G$

Case 1: $x \in H \implies xH = H = Hx$

Case 2: $x \in gH \implies x = gh$ for $h \in H \implies xH = ghH = gH$

Consider Hx since $g \notin H \implies Hg = gH \implies Hx = H(gh) = g(Hh) = gH = Hx$

Therefore $H \trianglelefteq G$ since $|G/H| = 2$ it is isomorphic to $\mathbb{Z}_2$

Example 1.9.4. Prove that the quotient group of a cyclic group is cyclic.

Let $G = \langle g \rangle$ be a group and $H \leq G$

Let $xH \in G/H$ s.t. $x \in G \implies x = g^k$ for $k \in \mathbb{Z}$

$\implies xH = g^kH = (gH)^k \implies G/H = \langle gH \rangle \implies$ it is cyclic

1.10 The Alternating Group

Definition 1.10.1. Two cycles are called transpositions

Definition 1.10.2. An element $\sigma \in S_n$ is called even(odd) if it can be written as the product of an even(odd) number of transpositions

Definition 1.10.3. The alternating group, A_n , is the subgroup of S_n made up of all even permutations.

$$A_n = \{\sigma \in S_n \mid \sigma \text{ is even}\}$$

Proposition 1.10.4. Every $\sigma \in S_n$ can be written as the product of transpositions.

Proof. Since σ can be written as the product disjoint cycles we only have to show that one cycles can be written as the product of transpositions.

Observe $(a_1, a_2, \dots, a_m) = (a_1, a_m)(a_1, a_{m-1}) \dots (a_1, a_2)$

A m cycle breaks into $m - 1$ transpositions, so an even length cycle is odd □

Proposition 1.10.5. If $(a_1, b_1)(a_2, b_2) \dots (a_n, b_n) = (1)$ then n is even, and thus $(1) \in A_n$

Observations:

$$(bc)(ab) = (ac)(bc) \tag{4}$$

$$(ac)(ab) = (ab)(bc) \tag{5}$$

$$(6)$$

Proof. Base Case: $n = 1 \implies (ab) \neq (1)$

$$n = 2 \implies (ab)(cd) = (1) \implies (ab) = (cd)$$

I.H. Assume for all $1 \leq k < m$ we have if $(a_1 b_1) \dots (a_k b_k) = (1)$

Assume $(a_1, b_1)(a_2, b_2) \dots (a_n, b_n) = (1)$

Set $a_m = a$ and use (1) or (2) to move the rightmost appearance of a to the left until

1. the two cycle containing a is next to its inverse

$$(a_1 b_1) \dots (ab_m) = (a_1 b_1) \dots (a_{j-1} b_{j-1})(ab)(ab)(a_{j+2} b_{j+1}) \dots (a_m b_m)$$

= a product of $m - 2$ transpositions by I.H. we have $m - 2$ is even $\implies m$ is even

2. the two cycle containing a never finds an inverse

$$(a_1 b_1) \dots (ab_m) = (ab)(a_2 b_2) \dots (a_m b_m) \implies a \text{ is not fixed} \implies \neq (1) \tag{6}$$

□

Proposition 1.10.6. A_n is a subgroup of order $\frac{n!}{2}$

Proof. $(1) \in A_n$ so nonempty Assume $\sigma, \mu \in A_n \implies \sigma = a_1 a_2 \dots a_r$ and $\mu = b_1 b_2 \dots b_s$ where r and s are even

$$\sigma \mu^{-1} = a_1 \dots a_r b_s \dots b_1 \quad r+s \text{ is even so } \sigma \mu^{-1} \in A_n$$

Thus, A_n is a subgroup

$$\text{Even} = \text{odd} \implies |A_n| = \frac{n!}{2} \quad \square$$

Corollary 1.10.7. $[S_n : A_n] = 2$ since $\# \text{even} = \# \text{odd} \implies A_n \trianglelefteq S_n \implies S_n/A_n \cong \mathbb{Z}_2$

Lemma 1.10.8. For $n \geq 3$, A_n is generated by 3-cycles.

Proof. Take $\sigma \in S_n$, write $\sigma = a_1 b_2 \dots a_m b_m$

$$a_j b_j = \begin{cases} (ab)(cd) = (acb)(acd) \\ (ab)(ac) = (acb) \end{cases} \implies \sigma \text{ can be written as 3-cycles} \quad \square$$

Lemma 1.10.9. Let $N \trianglelefteq A_n$ with $n \geq 3$. If N contains a 3-cycle then $N = A_n$

Proof. Note all 3-cycles can be written as products of $(12_)$. $a, b, c \in \{3, \dots, n\}$

$$\begin{aligned} (1a2) &= (12a)(12a) \\ (1ab) &= (12b)(12a)^2 \\ (2ab) &= (12b)^2(12a) \\ (abc) &= (12a)^2(12c)(12b)^2(12a) \end{aligned}$$

Take $(abc) \in N$

$$(2b)(1a)(abc)(1a)(2b) = (12c) \in N$$

Take $m \in \{3, \dots, n\}$, $(12m) = (12)(mc)(12c)^2(mc)(12) \in N \implies$ all 3-cycles are in N , but 3-cycles generate A_n so $N = A_n$ $\square$

Theorem 1.10.10. For $n \geq 5$, if $N \trianglelefteq A_n$ then $N = \{e\}$ or $N = A_n$

Proof. Take $(1) \neq \sigma \in N$ and write it as a product of disjoint cycles.

Case 1: There is a cycle of length ≥ 4

Write: $\sigma = (a_1 a_2 a_3 a_4 \dots a_m) \mu$

$(a_1 a_3 a_m) = \sigma^{-1}((a_1 a_2 a_3) \sigma (a_1 a_2 a_3)^{-1})$ since N is normal $\implies \in N \implies N = A_n$ by Lemma 1.10.8

Case 2: σ only contains 2,3-cycles

Case 2a: σ has at least two 3-cycles

$\sigma = (a_1 a_2 a_3)(a_4 a_5 a_6) \mu \implies (a_1 a_4 a_2 a_6 a_3) = \sigma^{-1}((a_1 a_2 a_4) \sigma (a_1 a_2 a_4)^{-1}) \in N$
 $\implies$ Case 1 $\implies N = A_n$

Case 2b: σ has one 3-cycle

$\sigma = (a_1 a_2 a_3) b_2 b_3 \dots b_k$

$\sigma^2 = (132) \in N \implies N = A_n$

Case 3: σ has only 2-cycles

$\sigma = (a_1 a_2)(a_3 a_4) \mu$

$(a_1 a_3)(a_2 a_4) = \sigma^{-1}((a_1 a_2 a_3) \sigma (a_1 a_2 a_3)^{-1}) \in N$

$(a_1 a_3 a_5) = (a_1 a_3)(a_2 a_4)((a_1 a_3 a_5)(a_1 a_3)(a_2 a_4)(a_1 a_3 a_5)^{-1}) \in N \implies N = A_n$

□

Definition 1.10.11. We say a group G is simple if it has no proper normal subgroup

1.10.1 Exercises

Example 1.10.12. $\mathbb{Z}_n$ is simple $\iff n$ is prime

Proof. Assume $\mathbb{Z}_n$ is simple

Since all subgroups of $\mathbb{Z}_n$ are normal $\implies$ the only subgroups of $\mathbb{Z}_n$ are $\mathbb{Z}_n$ and $\{e\}$

By Theorem 1.8.1 only $1, n \mid n \implies n$ is prime

Assume n is prime

Since all subgroups of $\mathbb{Z}_n$ are normal $\implies$ the only subgroups of $\mathbb{Z}_n$ are $\mathbb{Z}_n$ and $\{e\}$ since subgroups order have to divide n

Thus, $\mathbb{Z}_n$ is simple

□

1.11 External and Internal Direct Products

Definition 1.11.1. Given groups $(G_1, *_1), (G_2, *_2)$, their (external) direct product is the set

$$G_1 \times G_2 = \{(a, b) \mid a \in G_1, b \in G_2\}$$

with operation $*$:

$$(a, b) * (c, d) = (a *_1 c, b *_2 d)$$

Proposition 1.11.2. Suppose $a \in G_1, b \in G_2$ with $\text{ord}(a) = m$ and $\text{ord}(b) = n$ then $\text{ord}((a, b)) = \text{lcm}(m, n)$ in $G_1 \times G_2$

Proof. Set $k = \text{ord}((a, b)), l = \text{lcm}(m, n)$

$$\implies l = ms, l = nt \text{ for } s, t \in \mathbb{N}$$

$$(a, b)^l = (a^l, b^l) = ((a^m)^s, (b^n)^t) = (e_1, e_2) \implies k \mid l$$

$$(e_1, e_2) = (a, b)^k = (a^k, b^k) \implies a^k = e_1 \text{ and } b^k = e_2$$

$$\implies m \mid k \text{ and } n \mid k$$

$$\implies l \mid k \implies l = k$$

□

Corollary 1.11.3. $\text{ord}((g_1, g_2, \dots, g_n)) = \text{lcm}(\text{ord}(g_1), \dots, \text{ord}(g_n))$

Theorem 1.11.4. For $m, n \in \mathbb{N}$ we have

$$\mathbb{Z}_m \times \mathbb{Z}_n \cong \mathbb{Z}_{mn} \iff \gcd(m, n) = 1$$

Proof. ($\Leftarrow$) Assume the $\gcd(m, n) = 1$, take $(1, 1) \in \mathbb{Z}_m \times \mathbb{Z}_n$

$$\text{ord}((1, 1)) = \text{lcm}(\text{ord}(1), \text{ord}(1)) = \text{lcm}(m, n) = mn$$

$$\mathbb{Z}_{mn} \cong \langle (1, 1) \rangle \leq \mathbb{Z}_m \times \mathbb{Z}_n \text{ both have } mn \text{ elements thus } \mathbb{Z}_{mn} \cong \langle (1, 1) \rangle = \mathbb{Z}_m \times \mathbb{Z}_n$$

$$(\implies) \text{ Assume } \mathbb{Z}_{mn} \cong \mathbb{Z}_m \times \mathbb{Z}_n$$

Since $\mathbb{Z}_{mn}$ is cyclic, $\mathbb{Z}_m \times \mathbb{Z}_n$ is cyclic

$$\implies \exists (a, b) \in \mathbb{Z}_m \times \mathbb{Z}_n \text{ s.t. } \langle (a, b) \rangle = \mathbb{Z}_m \times \mathbb{Z}_n$$

$$mn = |\mathbb{Z}_{mn}| = |\mathbb{Z}_m \times \mathbb{Z}_n| = \text{ord}((a, b)) = \text{lcm}(\text{ord}(a), \text{ord}(b)) \leq \text{lcm}(m, n) = \frac{mn}{d}$$

$$\implies mn \leq \frac{mn}{d} \implies d \leq 1 \implies d = 1$$

□

Theorem 1.11.5. If $\gcd(m,n) = 1$ then $U_{mn} \cong U_m \times U_n$

Proof. $|U_{mn}| = \phi(mn) = \phi(m)\phi(n) = |U_m||U_n| = |U_m \times U_n|$

$$\psi : U_{mn} \rightarrow U_m \times U_n, \psi(a) = (a, a)$$

$$\text{Suppose in } U_{mn} a = b \implies a \equiv b \pmod{mn} \implies mn \mid b - a \implies m \mid b - a \text{ and } n \mid b - a$$

$$\implies a \equiv b \pmod{m} \text{ and } a \equiv b \pmod{n} \implies a = b \text{ in } U_n \text{ and } U_m$$

$$\implies \psi(a) = (a, a) = (b, b) = \psi(b) \implies \text{well-defined}$$

$$\text{Homomorphism: } \psi(ab) = (ab, ab) = (a, a)(b, b) = \psi(a)\psi(b)$$

$$\text{Injective: Let } a, b \in U_{mn} \text{ s.t. } \psi(a) = \psi(b)$$

$$\implies (a, a) = (b, b) \implies a \equiv b \pmod{n} \text{ and } a \equiv b \pmod{m}$$

$$\implies a \equiv b \pmod{mn} \implies a = b \quad \square$$

Proposition 1.11.6. If G is a group, $H \leq G$, and $N \trianglelefteq G$ then $HN = \{hn \mid h \in H, n \in N\} \leq G$

Proof. Let $x, y \in HN$

$$x = hn_1, y = jn_2$$

$$xy^{-1} = hn_1(jn_2)^{-1} = hn_1n_2^{-1}j^{-1} = hnj^{-1} = hj^{-1}(jnj^{-1}) = hn \in HN \quad \square$$

Definition 1.11.7. Suppose $N_1, N_2 \trianglelefteq G$. We say G is an internal direct product of N_1 and N_2 if

1. $N_1 \cap N_2 = \{e\}$
2. $G = N_1N_2$

Theorem 1.11.8. If G is an internal direct product of N_1 and N_2 then $G \cong N_1 \times N_2$

Proof. Define $\phi : N_1 \times N_2 \rightarrow G$

$$\phi(n_1, n_2) = n_1n_2$$

$$\text{Homomorphism: Let } (a, b), (c, d) \in N_1 \times N_2$$

$$\phi((a, b)(c, d)) = \phi((ac, bd)) = acbd$$

$$\text{Consider: } cbc^{-1}b^{-1} \in N_2, N_1 \implies = e \implies cb = bc$$

$$\implies = abcd = \phi(a, b)\phi(c, d)$$

$$\text{Injective: } \phi(a, b) = \phi(c, d)$$

$$\implies ab = cd \implies c^{-1}a = db^{-1} \in N_2, N_2 \implies = e$$

$$\implies c = a \text{ and } b = d$$

$$\text{Surjective: Take } y \in G \implies y = ab, a \in N_1 \text{ and } b \in N_2 \quad \phi(ab) = ab = y \quad \square$$

1.12 First Isomorphism Theorem of Groups

Proposition 1.12.1. *If $\varphi : G_1 \rightarrow G_2$ is a homomorphism, $H \leq G_1$, $K \leq G_2$, and $N \trianglelefteq G_2$ then*

1. $\varphi(H) \leq G_2$

2. $\varphi(K)^{-1} \leq G_1$

3. $\varphi(N) \trianglelefteq G_1$

1. *Proof.* Let $x, y \in \varphi(H)$

$$\implies a, b \in H \text{ s.t. } x = \varphi(a), y = \varphi(b)$$

$$xy^{-1} = \varphi(a)(\varphi(b))^{-1} = \varphi(ab^{-1}) \in \varphi(H) \implies \varphi(H) \leq G_2 \quad \square$$

2. *Proof.* Let $a, b \in \varphi(K)^{-1}$

$$\implies \varphi(a), \varphi(b) \in K$$

$$\varphi(a)\varphi(b)^{-1} \in K \implies \varphi(ab^{-1}) \in K$$

$$ab^{-1} \in \varphi(K)^{-1} \implies \varphi(K)^{-1} \leq G_1 \quad \square$$

3. *Proof.* Let $n \in \varphi(N)^{-1}$ and $g \in G_1$

$$\implies \varphi(n) \in N, \varphi(g) \in G_2$$

$$\implies \varphi(g)\varphi(n)\varphi(g)^{-1} \in N \implies \varphi(gng^{-1}) \in N$$

$$\implies gng^{-1} \in \varphi(N)^{-1} \quad \square$$

Theorem 1.12.2. Suppose $\varphi : G \rightarrow G_2$ is a group homomorphism then there is a unique isomorphism $\psi : G/\ker\varphi \rightarrow \text{im}\varphi$ s.t. the diagram commutes

$$\begin{array}{ccc} G & \xrightarrow{\varphi} & \text{im}(\varphi) \leq G_2 \\ \downarrow \pi & \nearrow \psi & \\ G/\ker\varphi & & \end{array}$$

Commutes: $\psi(g \ker\varphi) = \varphi(g)$

Proof. ψ is well-defined:

$$\begin{aligned} \text{Let } x\ker\varphi = y\ker\varphi \in G/\ker\varphi &\implies xy^{-1} \in \ker\varphi \\ \implies \varphi(xy^{-1}) = e &\implies \varphi(x) = \varphi(y) = \psi(y\ker\varphi) \end{aligned}$$

Homomorphism:

$$\psi((x\ker\varphi)(y\ker\varphi)) = \psi((xy)\ker\varphi) = \varphi(xy) = \varphi(x)\varphi(y) = \psi(x\ker\varphi)\psi(y\ker\varphi)$$

Injective:

$$\begin{aligned} \text{Let } g\ker\varphi \in \ker\psi \\ \implies \psi(g\ker\varphi) = e &\implies \varphi(g) = e \implies g \in \ker\varphi \\ \implies g\ker\varphi = \ker\varphi &\implies \psi \text{ is Injective} \end{aligned}$$

Surjective:

$$\begin{aligned} \text{Let } y \in \text{im}\varphi &\implies \exists g \in G \text{ s.t. } \varphi(g) = y. \\ \implies \psi(g\ker\varphi) = \varphi(g) &= y \end{aligned}$$

□

1.12.1 Exercises

Example 1.12.3. Given a group G , an inner automorphism is a function $\psi_g : G \rightarrow G$ s.t. $\psi_g(x) = gxg^{-1}$

1. Show that ψ_g is an isomorphism
2. Show $\text{Inn}(G) = \{\psi_g \mid g \in G\}$ is a group.
3. Use the first isomorphism theorem to show

$$G/Z(G) \cong \text{Inn}(G)$$

Proof. 1. Injective:

Let $g \in G$ and $y \in \ker(\psi_g)$

$$\implies y = gxg^{-1} = e \text{ s.t. } x \in G$$

$$\implies x = e \implies y = e \implies \psi_g \text{ is bijective since } |G| = |G|$$

Homomorphism:

Let $a, b \in G$

$$\psi_g(ab) = gabg^{-1} = gag^{-1}gbg^{-1} = \psi_g(a)\psi_g(b)$$

2. Group

(a) Closure ✓

$$(b) \text{ Identity } \psi_e \circ \psi_g(n) = gng^{-1} = \psi_g \circ \psi_e(n) = \psi_g(n)$$

$$(c) \text{ Inverses } \psi_g \circ \psi_{g^{-1}}(n) = gg^{-1}ngg^{-1} = n = \psi_{g^{-1}} \circ \psi_g(n) = \psi_e(n)$$

(d) Associativity ✓

3. By Lemma 1.4.5 $Z(G) \leq G$

Normal:

$$gZ = Zg \implies Z(G) \trianglelefteq G$$

$$\implies G/Z(G) \text{ is a group}$$

Define: $\phi : G \rightarrow \text{Inn}(G)$, $\phi(g) = \psi_g$

$\text{Im}(\phi) = \text{Inn}(G)$ by definition

Consider: $\ker(\phi)$, $Z(G)$

$$\text{Let } x \in \ker(\phi) \implies \psi_x(g) = g \implies xgx^{-1} = g \implies xg = gx \implies x \in Z(G)$$

Let $x \in Z(G) \implies \varphi(x) = \psi_x(g) = xgx^{-1} = g \implies x \in \ker(\varphi)$

$\implies Z(G) = \ker(\varphi)$

By Theorem [1.12.2](#), $G/Z(G) \cong \text{Inn}(G)$

□

1.13 More Isomorphism Theorems

Theorem 1.13.1 (Second Isomorphism Theorem). *Suppose $H \leq G$ and $N \trianglelefteq G$ then $H \cap N \trianglelefteq H$ and $H/H \cap N \cong HN/N$*

Proof. Suppose $h \in H$, $n \in H \cap N$

Since $N \trianglelefteq G$ and $h \in G$

$$\implies hnh^{-1} \in N$$

Since $h, n \in H \implies hnh^{-1} \in H \implies hnh^{-1} \in H \cap N$

$$\implies H \cap N \trianglelefteq H$$

Define : $\varphi : H \rightarrow HN/N$

$$h \mapsto hN$$

Homomorphism: $\varphi(xy) = (xy)N = xNyN = \varphi(x)\varphi(y)$

Surjective: Take $(hn)N \in HN/N$, for $h \in H$, $n \in N$

$(hn)N = hN$ because $h^{-1}(hn) = n \in N$

$$\varphi(h) = hN = (hn)N$$

Kernel: Let $h \in \ker(H) \trianglelefteq H \implies h \in H$

$$\varphi(h) = hN = N \implies h \in N \implies h \in H \cap N \implies \ker(H) = H \cap N$$

By Theorem 1.12.2 $H/H \cap N \cong HN/N$

□

Theorem 1.13.2 (Third Isomorphism Theorem). *If $M, N \trianglelefteq G$ where $N \leq M$ then*

$$(G/N)/(M/N) \cong G/M$$

Proof. Define: $\varphi : G/N \rightarrow G/M$

$$gN \mapsto gM$$

Well Defined: Suppose $xN = yN \in G/N \implies xy^{-1} \in N \leq M$

$$\implies \varphi(xN) = xM = yM = \varphi(yN)$$

Homomorphism: $\varphi(xNyN) = \varphi(xyN) = xyM = xMyM = \varphi(xN)\varphi(yN)$

Surjective: Suppose $gM \in G/M \implies \varphi(gN) = gM$

Kernel: Suppose $gN \in \ker(\varphi) \iff \varphi(gN) = M = gM \iff g \in M \iff gN \in M/N$

$$\ker(\varphi) = M/N$$

By Theorem 1.12.2 $(G/N)/(M/N) \cong G/M$

□

Corollary 1.13.3. $\mathbb{Z}_n / \langle m \rangle \cong \mathbb{Z}_{\gcd(m,n)}$

Proof. Set $d = \gcd(m, n)$ by Theorem 1.5.9 $\langle m \rangle = \langle d \rangle \implies \text{ord} = \frac{n}{d}$
 $\implies \cong \mathbb{Z}_{\frac{n}{d}} \cong d\mathbb{Z} / n\mathbb{Z}$
 $\mathbb{Z}_n / \langle m \rangle \cong \mathbb{Z} / n\mathbb{Z} / d\mathbb{Z} / n\mathbb{Z} \cong \mathbb{Z} / d\mathbb{Z} \cong \mathbb{Z}_d$ □

Theorem 1.13.4 (Fourth Isomorphism Theorem). *Suppose $N \trianglelefteq G$. Every subgroup $K \leq G/N$ is the form $K = H/N$ where $N \leq H \leq G$. Further there is a bijection $\{H \leq G \mid N \leq H\} \longleftrightarrow \{K \leq G/N\}$*

Proof. Suppose $K \leq G/N$. Take $K = \{gN \mid g \in H\}$ s.t. $H = \{g \in G \mid gN \in K\}$
 $e \in H$ b/c $eN = N \in K$ b/c $K \leq G/N$
 Suppose $x, y \in H \implies xN, yN \in K \implies xN(yN)^{-1} \in K \implies xy^{-1}N \in K \implies xy^{-1} \in H$
 $\implies H \leq G$
 Consider the projection:

$$\begin{aligned} \Pi : G &\rightarrow G/N \\ g &\mapsto gN \end{aligned}$$

Define F by $F(H) = \Pi(H) = H/N$

Injective: Assume $F(H_1) = F(H_2)$

$$\implies H_1/N = H_2/N$$

Suppose $h \in H_1 \iff hN \in H_1/N = H_2/N \iff h \in H_2$

$$\implies H_1 = H_2$$

Surjective: Suppose $K \leq G/N \implies K = H/N$ s.t. $N \leq H \leq G$

$$F(H) = H/N = K$$

□

1.13.1 Exercises

Example 1.13.5. If $M, N \trianglelefteq G$ s.t. $G = MN$ then $G/M \cap N \cong G/M \times G/N$

Proof. Assume $M, N \trianglelefteq G$ s.t. $G = MN$

Let $\varphi(g) = (gM, gN)$

Surjective: Let $(aM, bN) \in G/M \times G/N$ s.t. $a, b \in G$

Since $G = MN \exists g \in G$ s.t. $g = ab \implies \varphi(g) = (gM, gN) = (aM, bN)$

Kernel: Let $g \in \ker(\varphi) \iff \varphi(g) = (M, N) = (gM, gN)$

$$\iff M = gM \text{ and } N = gN \iff g \in M \cap N$$

By Theorem 1.12.2 $G/M \cap N \cong (G/M) \times (G/N)$ □

1.14 Fundamental Theorem of Abelian Groups

Theorem 1.14.1 (Fundamental Theorem of Finite Abelian Groups). *If G is a finite abelian group then*

$$G \cong \mathbb{Z}_{p_1^{a_1}} \times \mathbb{Z}_{p_2^{a_2}} \times \cdots \times \mathbb{Z}_{p_n^{a_n}}$$

for (not necessarily distinct) primes p_i and $a_i \in \mathbb{N}$

Lemma 1.14.2. *Let G be an abelian group with $|G| = n$ and p a prime s.t. $p \mid n$ then G has an element of order p*

Proof. Base Case: $n = 1$

$$G = \{e\}$$

IH: Assume for all $1 \leq m < n$ the statement holds

Consider: $|G| = n$

Case 1: G has no proper subgroups. If $e \neq g \in G$

$$\{e\} \leq \langle g^{\frac{n}{p}} \rangle \leq \langle g \rangle = G$$

$$\text{Case 1a: } \{e\} = \langle g^{\frac{n}{p}} \rangle \implies e = g^{\frac{n}{p}} \implies \frac{n}{p} \text{ is a multiple of } n \implies p = 1 \implies \Leftarrow$$

$$\text{Case 1b: } \langle g^{\frac{n}{p}} \rangle = \langle g \rangle \implies p = \text{ord}(g^{\frac{n}{p}}) = \text{ord}(g) = n \implies g \text{ is order } p$$

Case 2: G has a proper subgroup $H \leq G$ s.t. $1 < |H| < n$,

also $H \trianglelefteq G$ since G is an abelian group.

Case 2a: $p \mid |H|$ then by IH we have $h \in H$ where $\text{ord}(h) = p$

Case 2b: $p \nmid |H|$ then $p \mid |G/H|$ b/c $n = |G| = |H||G/H| \Leftarrow$ Theorem 1.8.1

Apply IH to G/H $gH \in G/H$ where $\text{ord}(gH) = p$

$$\implies gH \neq H \implies g \notin H$$

$$H = (gH)^p = g^p H \implies g^p \in H. \text{ Set } |H| = m \implies (g^p)^m = e \implies (g^m)^p = e$$

Assume $\text{ord}(g^m) \neq p \implies \text{ord}(g^m) = 1 \implies g^m = e$, note $\text{gcd}(m, p) = 1$ b/c $p \nmid m$

$$\implies mx + py = 1 \text{ for } x, y \in \mathbb{Z}$$

$$g = (g^m)^x (g^p)^y \implies g = eh \text{ for some } h \in H \implies \Leftarrow$$

$$\text{ord}(g^m) = p$$

□

Definition 1.14.3. We say G is a p -group if for all $g \in G$ $\text{ord}(g) = p^a$ for some $a \in \mathbb{Z}_{\geq 0}$

Lemma 1.14.4. G is a finite abelian p -group $\iff |G| = p^n$

Proof.

($\implies$) Assume p, q are distinct primes s.t. $p \mid |G|$ and $q \mid |G|$

By Lemma 1.14.2 find $x, y \in G$ s.t. $\text{ord}(x) = p$ and $\text{ord}(y) = q$

$\implies G$ is not a p -group

($\impliedby$) Assume $|G| = p^n$ and $g \in G$

$\langle g \rangle \leq G \implies |\langle g \rangle| \mid |G| \implies \text{ord}(g) \mid p^n \implies \text{ord}(g) = p^m$ for $0 \leq m \leq n$ □

Lemma 1.14.5. Let G be a finite abelian group of order $p_1^{r_1} p_2^{r_2} \cdots p_k^{r_k}$ for distinct primes p_i then

$$G \cong G_1 \times G_2 \times \cdots \times G_k$$

with $|G_i| = p_i^{r_i}$

Proof. Set $G_i = \{g \in G \mid \text{ord}(g) = p_i^n \text{ s.t. } n \in \mathbb{N}\}$

Subgroups: $e \in G_i, \text{ord}(e) = 1 = p_i^0$

Closure: Let $x, y \in G_i \implies \text{ord}(x) = p_i^m, \text{ord}(y) = p_i^n$

$\text{ord}(xy) = \text{lcm}(p_i^m, p_i^n) = p_i^{\max(m, n)} \implies xy \in G_i$

Inverses: Let $x \in G_i \implies \text{ord}(x) = p_i^n \implies \text{ord}(x^{-1}) = p_i^n$

$\implies G_i \leq G$

Let $g \in G_i \cap G_j$ s.t. $i \neq j \implies \text{ord}(g) = p_i^m = p_j^n \implies m, n = 0$ since these are distinct primes

$\implies g = e \implies G_i \cap G_j = \{e\}$

Let $g \in G \implies \text{ord}(g) \mid p_1^{r_1} \cdots p_k^{r_k}$

Set $a_i = p_1^{r_1} \cdots p_{i-1}^{r_{i-1}} p_{i+1}^{r_{i+1}} \cdots p_k^{r_k} = \frac{|G|}{p_i^{r_i}}$

$\text{gcd}(a_1, a_2, \dots, a_k) = 1$

$\implies$ find $b_i \in \mathbb{Z}$ s.t. $a_1 b_1 + \cdots + a_k b_k = 1$

$g = g^{a_1 b_1} \cdots g^{a_k b_k} = g_1 \cdots g_k$ where $g_i = g^{a_i b_i}$

Note: $g_i^{p_i^{r_i}} = g^{(a_i p_i^{r_i}) b_i} = g^{|G| b_i} = e \implies g_i \in G_i \implies G = G_1 G_2 \cdots G_k$

By Theorem 1.11.8, $G \cong G_1 \times \cdots \times G_k$ □

Lemma 1.14.6. *Let G be a finite abelian p -group and take $g \in G$ with maximal order, then*

$$G \cong \langle g \rangle \times K$$

for some $K \leq G$

Proof. Induction on n , where $|G| = p^n$

Base Case: Let $n = 1$, $G \cong \mathbb{Z}_p \cong \langle g \rangle \times \{e\}$

IH: Assume for $1 \leq m < n$ the statement holds

Consider $|G| = p^n$ and $g \in G$ s.t. $\text{ord}(g)$ is maximal

$$\implies \text{ord}(g) = p^k$$

If $k = n$ $\text{ord}(g) = \text{ord}(G) \implies G = \langle g \rangle \cong \langle g \rangle \times \{e\}$

If $k < n$, choose $h \in G \setminus \langle g \rangle$ of min order

$$\text{ord}(h) = p^r \leq p^k \implies \text{ord}(h^p) = p^{r-1} < \text{ord}(h)$$

$$\implies h^p \in \langle g \rangle \implies h^p = g^a \text{ for } a \geq 0$$

$$(g^a)^{p^{k-1}} = (h^p)^{p^{k-1}} = h^{p^k} = e \text{ since } p^{r-1} \mid p^k$$

$$\implies \text{ord}(g^a) = \frac{\text{ord}(g)}{\gcd(\text{ord}(g), a)} \leq p^{k-1} \implies p \mid a$$

$$\implies a = ps \implies h^p = g^{ps} \implies (g^{-s}h)^p = e \text{ set } x = g^{-s}h \neq e$$

$$\implies x^p = e \implies \text{ord}(x) = p \implies \text{ord}(h) = p$$

Take $y \in \langle g \rangle \cap \langle h \rangle \implies y = h^a, 0 \leq a \leq p-1$

If $a \neq 0$ then $\gcd(a, p) = 1$ then take $r, s \in \mathbb{Z}$ s.t.

$$ar + ps = 1. \text{ Now } y^r = (h^a)^r = h^{ar} = h^{1-ps} = h(h^p)^{-s} = h \implies h \in \langle g \rangle \Rightarrow \Leftarrow$$

$$\implies a = 0 \implies y = e \implies \langle g \rangle \cap \langle h \rangle = \{e\}$$

Take $H = \langle h \rangle$

Let $gH \in G/H$ and $|G/H| = p^{n-1}$

Set $r = \text{ord}(gH) \mid \text{ord}(g)$. Also $g^r H = (gH)^r = H \implies g^r \in H \implies g^r \in \langle g \rangle \cap \langle h \rangle$

$$\implies g^r = e \implies \text{ord}(g) \mid r \implies r = \text{ord}(g) = p^k$$

Since $\text{ord}(xH) \mid \text{ord}(x) \implies \text{ord}(gH)$ is maximal

Apply IH on $G/H \implies G/H \cong \langle gH \rangle \times J$

where by Theorem 1.13.4 $J = K/H$ s.t. $H \leq K \leq G$

Let $x \in \langle g \rangle \cap K \implies xH \in \langle gH \rangle \cap K/H \implies xH = eH \implies x \in H \implies x = e$

$$\implies \langle g \rangle \cap K = \{e\}$$

Let $x \in G \implies xH \in G/H = \langle gH \rangle (K/H)$

$$\implies xH = g^r KH \implies x \in g^r KH \subseteq \langle g \rangle K \text{ since } H \leq K$$

By Theorem 1.11.8, $G \cong \langle g \rangle \times K$

□

Fundamental Theorem of Finite Abelian Groups

Theorem 1.14.1

Proof. Assume G is a finite abelian group

By Lemma 1.14.5, $G \cong G_1 \times G_2 \times \cdots \times G_k$ where $|G_i| = p_i^{r_i}$

By Lemma 1.14.6, $G_i = \langle g_i^{(1)} \rangle \times H_i$ where $\text{ord}(g_i^{(1)}) = p_i^{s_i^{(1)}}$

By Theorem 1.6.2, $G_i \cong \mathbb{Z}_{p_i^{s_i^{(1)}}} \times H_i$

Then repeat this on H_i until $G_i \cong \mathbb{Z}_{p_i^{s_i^{(1)}}} \times \cdots \times \mathbb{Z}_{p_i^{s_i^{(m)}}}$

$\implies G \cong \mathbb{Z}_{p_1^{a_1}} \times \mathbb{Z}_{p_2^{a_2}} \times \cdots \times \mathbb{Z}_{p_n^{a_n}}$

□

Example 1.14.7. All abelian groups of order $8 = 2^3$.

$$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$$

$$\mathbb{Z}_2 \times \mathbb{Z}_4$$

$$\mathbb{Z}_8$$

1.14.1 Exercises

Example 1.14.8. All abelian groups of order $720 = 2^4 3^2 5$

$$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_5$$

$$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_4 \times \mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_5$$

$$\mathbb{Z}_2 \times \mathbb{Z}_8 \times \mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_5$$

$$\mathbb{Z}_{16} \times \mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_5$$

$$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_9 \times \mathbb{Z}_5$$

$$\mathbb{Z}_2 \times \mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_9 \times \mathbb{Z}_5$$

$$\mathbb{Z}_2 \times \mathbb{Z}_8 \times \mathbb{Z}_9 \times \mathbb{Z}_5$$

$$\mathbb{Z}_{16} \times \mathbb{Z}_9 \times \mathbb{Z}_5$$

$$\mathbb{Z}_4 \times \mathbb{Z}_4 \times \mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_5$$

$$\mathbb{Z}_4 \times \mathbb{Z}_4 \times \mathbb{Z}_9 \times \mathbb{Z}_5$$

Example 1.14.9. Show that $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \cdots$ is not finitely generated.

Assume it is finitely generated.

Everything has order 2 $\implies G = \mathbb{Z}_2 \times \mathbb{Z}_2 \times \cdots \cong \mathbb{Z}_2 \times \mathbb{Z}_2 \times \cdots \mathbb{Z}_2 = (\mathbb{Z}_2)^n$

$G \cong (\mathbb{Z}_2)^n \times G, \{e\} \times G \trianglelefteq (\mathbb{Z}_2)^n \times G \cong G$

$G/G \cong \{e\} \cong (\mathbb{Z}_2)^n \cong (\mathbb{Z}_2)^n \times G/\{e\} \times G$

$\implies G$ has order 1. $\implies \Leftarrow$